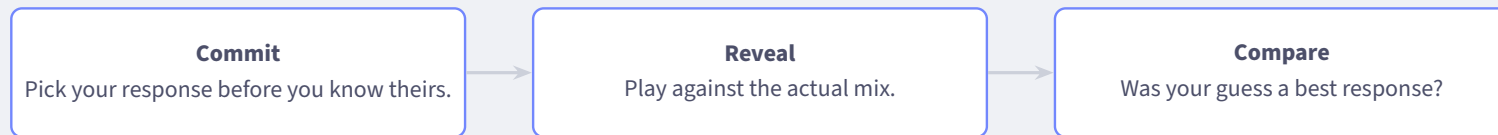


Rationalisation

Guess the mix, commit, then find out



You are the **column** player throughout.

$$A = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$$

The row player will mix. You have to choose c_1 or c_2 before you know how.

The row player's mix	You guessed	Best response	Your payoff
$\sigma_1 = (0.2, 0.8)$			
$\sigma_1 = (0.9, 0.1)$			
$\sigma_1 = (1/3, 2/3)$			

How many pairs guessed right before the reveal?

Against $\sigma_1 = (p, 1 - p)$ your two actions pay

$$u_2(c_1, \sigma_1) =$$

$$u_2(c_2, \sigma_1) =$$

So c_1 is the better reply exactly when p _____, and the two are equal at $p =$ _____.

Is c_1 dominated by c_2 ?

Is c_2 dominated by c_1 ?

Neither, because which one is better depends on _____.

Dominance only settles an action that is worse against *every* strategy of the opponent. Here each action is a best response to some mixes, so we need the _____ instead.

.....

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At $\sigma_1 = (1/3, 2/3)$ both of your actions pay , so every strategy of yours is a best response.

That is the row player's half of the equilibrium: they mix precisely so that you cannot exploit them. Which of your own mixes does the same to *them*?

Dominance and best responses are not examined on their own. They are examined together with **Nash Equilibrium**, so the natural follow-up is **Question 1** on that page, where the best response condition does exactly this job for Rock-Paper-Scissors.

The habit worth building now: before computing anything, ask which actions could ever be a best response. Anything that never can be is not worth carrying into the algebra.

vknights.org/gt/topics/nash-equilibrium.html